

N01: Single Phase Transformer

Construction, Working Principle, and Losses

Built to be EXPLAINED in the paper, not just recited. Every part has a function, every function has a one-line reason, every graph has a label checklist.

QUESTION SHAPES THIS FILE ANSWERS

A1-01: Explain the construction and working of a single phase transformer.

A1-20: List the losses in a transformer and explain how to reduce them.

Honest note: understanding beats rereading. After each slide, close it and say the explanation out loud once.

Question shape and the mark grammar

READ THIS FIRST

A1-01 (the big one):

Explain the construction and working of a single phase transformer.

A1-20:

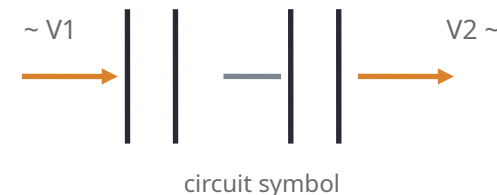
What are the losses in a transformer? Explain the methods to reduce them.

WHAT TO WRITE (mark components, in order)

- 1. One-line definition (transformer = static machine, mutual induction)
- 2. The construction diagram, fully labeled (diagram marks)
- 3. Each part + its function in one line (2 marks live here)
- 4. Working: the 6-step flux story + Faraday's law named
- 5. One-line conclusion (voltage transforms, frequency does not)

KEY DISTINCTION A transformer has NO rotating part and NO air gap: it is a static device. So it has no mechanical (friction/windage) losses at all. Its entire loss story is copper + iron. A DC machine problem's 'mechanical losses' never apply here.

One-line conclusion to memorize: definition, labeled diagram, part functions, Faraday, conclusion. In that order; every time.



Construction: the parts and their jobs

ROTE 1 | CONSTRUCTION

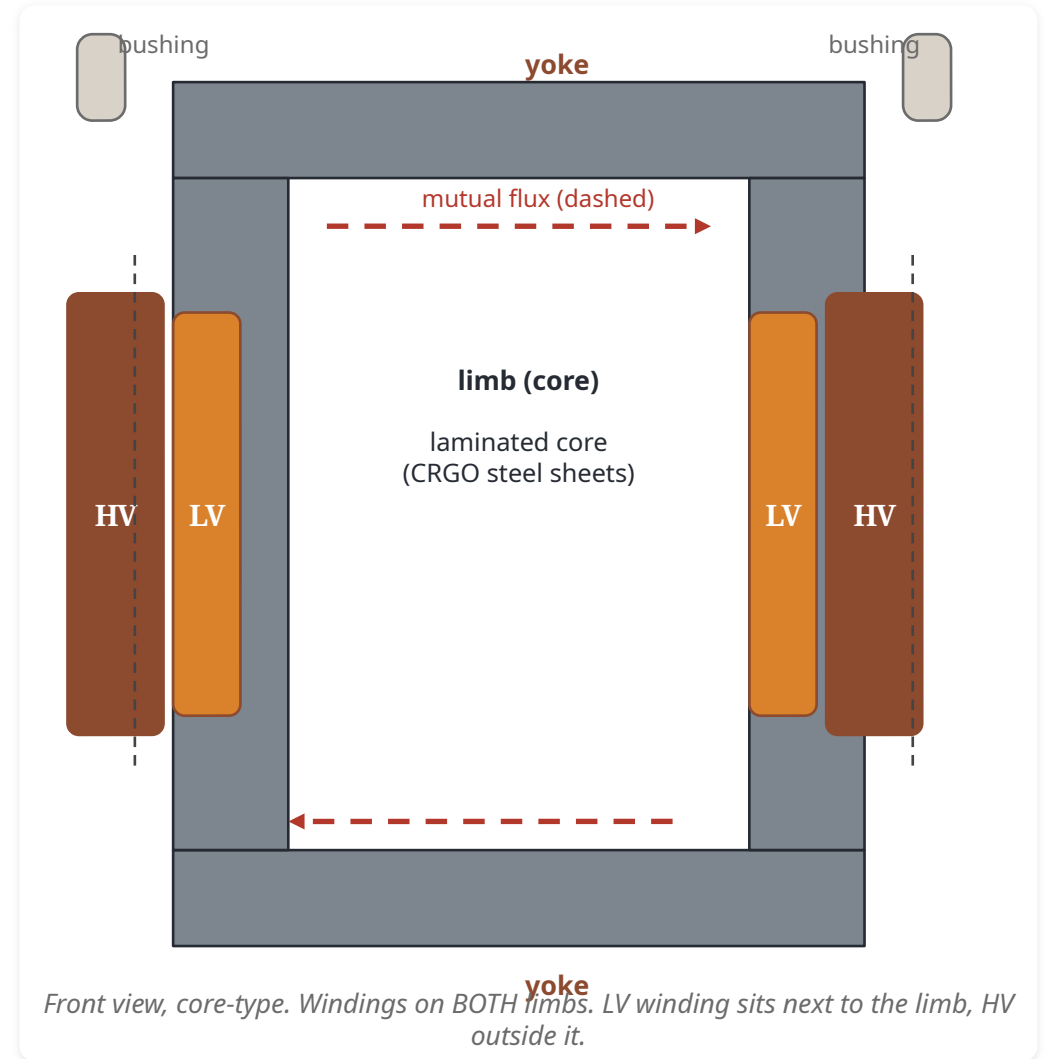
Magnetic core + yoke Laminated CRGO (cold-rolled grain-oriented) steel. The limbs carry the windings; the yoke closes the flux path so flux has one low-reluctance loop. Also the mechanical frame.

Windings (LV + HV) Primary and secondary, insulated copper. LV: fewer turns of thick wire. HV: more turns of thin wire. Wound concentrically: LV nearest the limb, HV over it (insulation cost is lowest this way).

Insulation Enamel on conductors, pressboard/paper between windings and core, transformer oil in the tank. The HV winding sits OUTSIDE because that is where the biggest insulation distance to earth is cheapest.

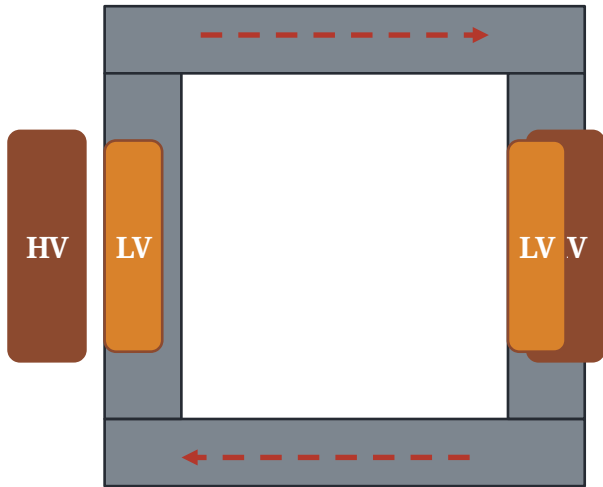
Tank, oil, accessories Sheet-steel tank with transformer oil: cools and insulates. Conservator takes up oil expansion. Bushings bring terminals out of the tank. Silica-gel breather dries the air. These add up to a large share of total weight.

One-line conclusion: core gives the flux path, windings give the voltage, insulation and oil keep them apart and cool.



Core type vs shell type

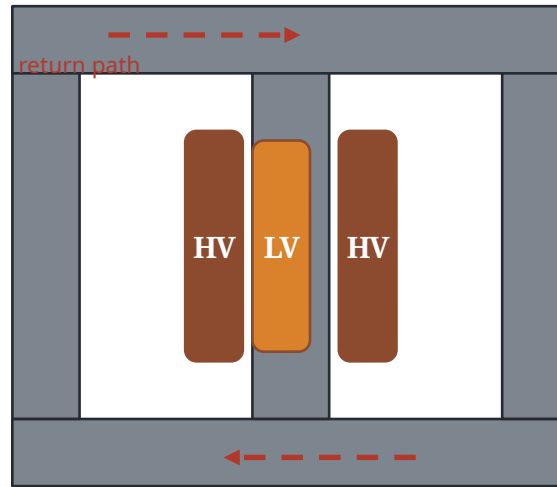
ROTE 1 | TWO CONSTRUCTIONS



CORE TYPE

winding surrounds the core. Windings on **BOTH** limbs.

One-line conclusion: core type = winding around core (HV-friendly). Shell type = core around winding (LV high-current, mechanically protected).



SHELL TYPE

core surrounds the winding. Windings only on the **CENTER** limb; flux splits into two return paths.

QUICK COMPARE

Flux path

core: one loop through 2 limbs. shell: splits through 2 outer limbs.

Windings

core: concentric, on both limbs. shell: sandwich, on center limb only.

Insulation

core type: easier HV insulation, more insulation volume. shell: windings shielded by outer limbs.

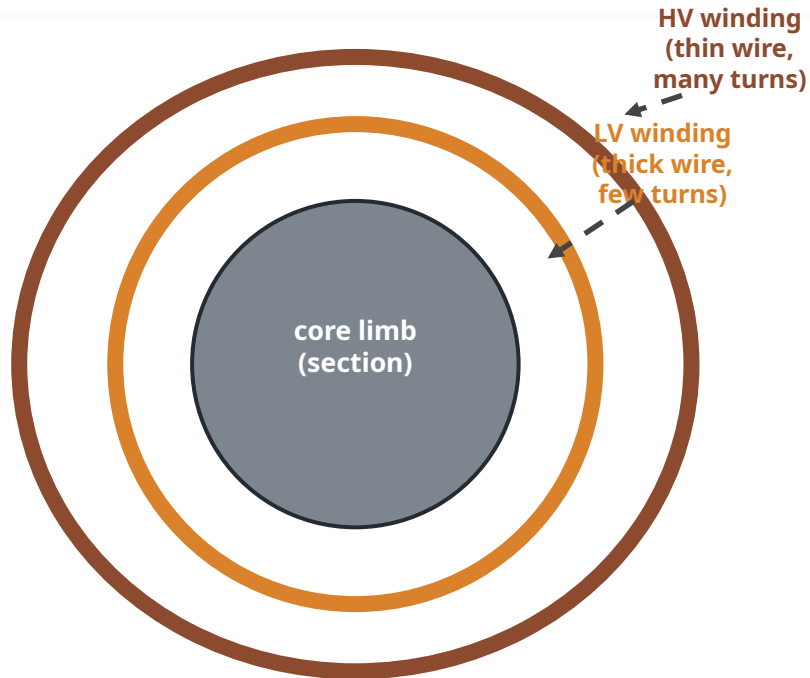
Used for

core type: high voltage, power transformers. shell type: low voltage, high current, small transformers.

Draw labels the examiner wants: number of limbs, where the windings sit, and the flux path direction.

Windings: why LV is inside and HV is outside

ROTE 1 | WINDING DETAIL



insulation between every layer, and to earth

Limb cross-section (concentric windings)

LV winding: few turns of THICK wire (high current, low voltage). Placed next to the limb.

HV winding: many turns of THIN wire (low current, high voltage). Placed over the LV winding.

WHY this order (the explain-the-reason marks)

The core is at earth potential. The LV winding has a small voltage, so it needs only thin insulation to the core. The HV winding is then insulated from the LV, and since HV sits outside, its insulation to the tank/earth has the longest path. Total insulation volume is minimum this way.

COIL TYPES

Concentric: cylindrical, LV inside HV (core type). Sandwich: discs of LV and HV stacked alternately (shell type).

URNS LAW (preview of N02)

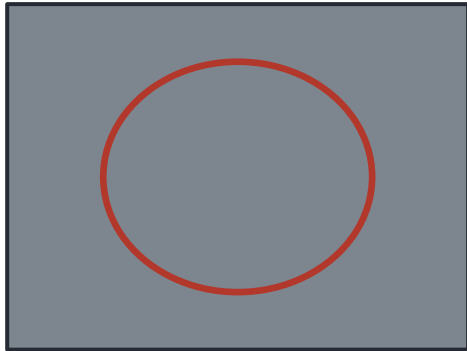
$E1/E2 = N1/N2 = a$. Voltage transforms with turns, frequency never changes.

One-line conclusion: LV thick-and-few inside, HV thin-and-many outside: minimum insulation, same flux.

Why laminated? (the eddy-current story)

ROTE 1 | THE WHY LAYER

big loops = big I^2R heating



SOLID core: large eddy-current loops free to circulate

insulation between sheets blocks the loops



LAMINATED core: loops trapped inside one thin sheet (thickness t)

THE TWO IRON-LOSS FORMULAS

$$P_{\text{hysteresis}} = k_h \times f \times B_m^{1.6}$$

cut it with better steel: CRGO / silicon steel has a narrow B-H loop, so less energy lost per cycle.

$$P_{\text{eddy}} = k_e \times f^2 \times B_m^2 \times t^2$$

cut it by laminating: t = sheet thickness, and the loss falls with t SQUARED. 0.27-0.35 mm sheets, insulated from each other.

Why CRGO steel?

Narrow hysteresis loop: less energy dissipated per magnetisation cycle.

Why a yoke?

Closes the magnetic circuit so the flux has one continuous low-reluctance path; also mechanically braces the limbs.

Why oil + conservator?

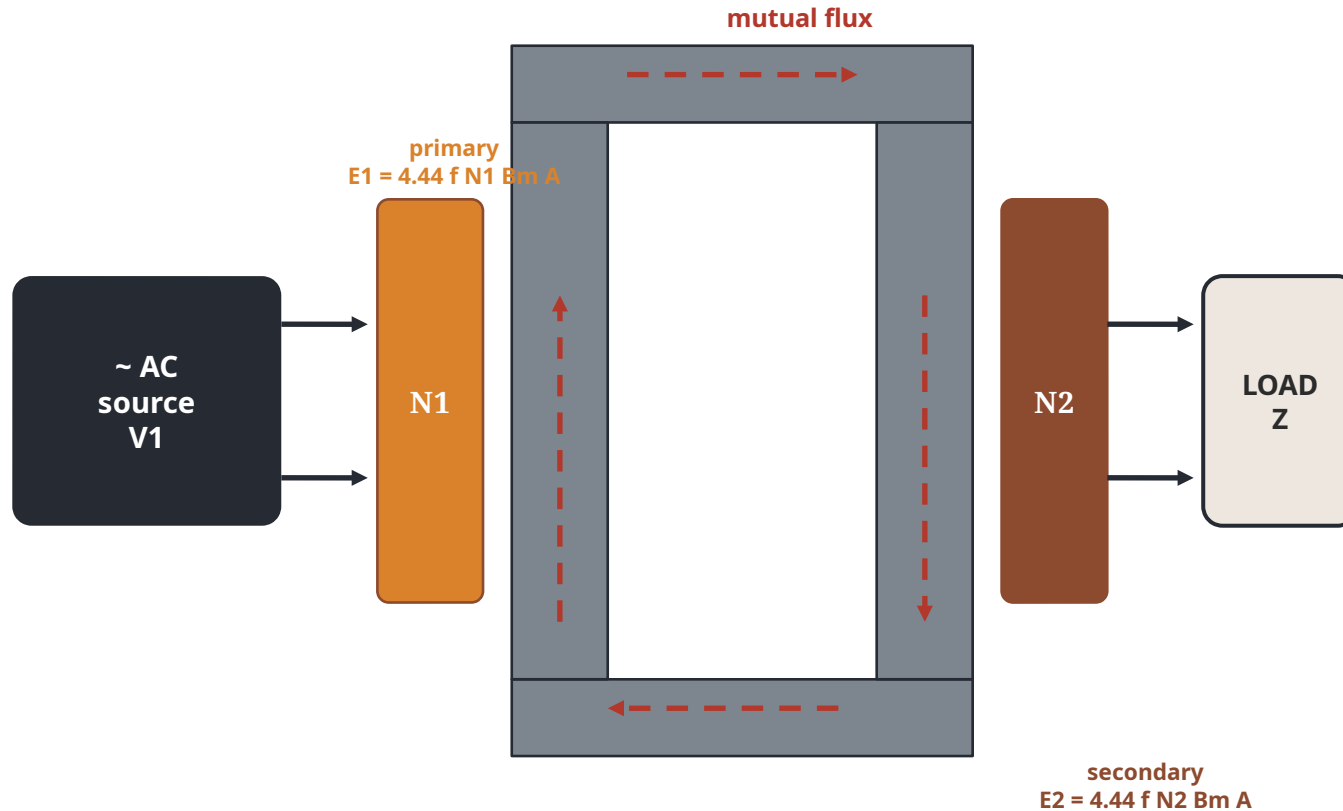
Oil insulates and cools; the conservator gives the oil room to expand so the tank never over-pressures.

Why a breather?

Silica gel dries the air entering as oil volume changes: moisture is insulation's enemy.

Working principle: the six-step flux story

ROTE 2 | WORKING



Working diagram: AC in N1 -> alternating flux in the core -> flux links N2 -> emf induced in N2 by Faraday -> load current.

One-line conclusion: one alternating flux, two circuits. Power crosses by mutual induction at UNCHANGED frequency.

THE SEQUENCE TO WRITE

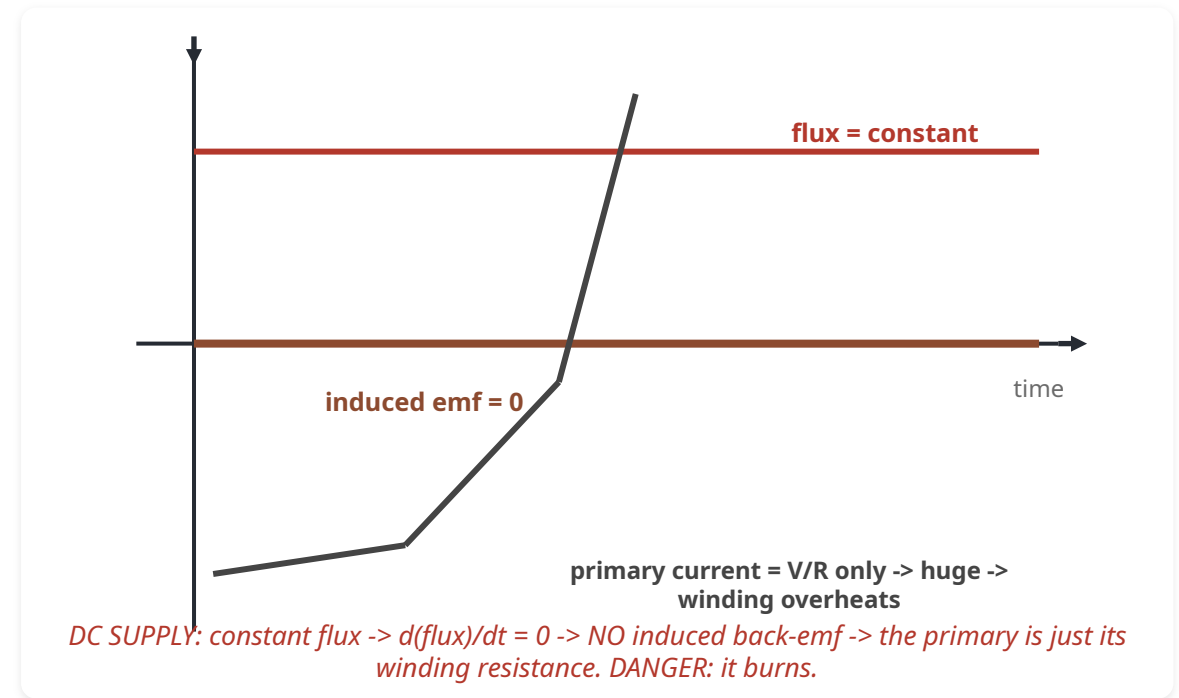
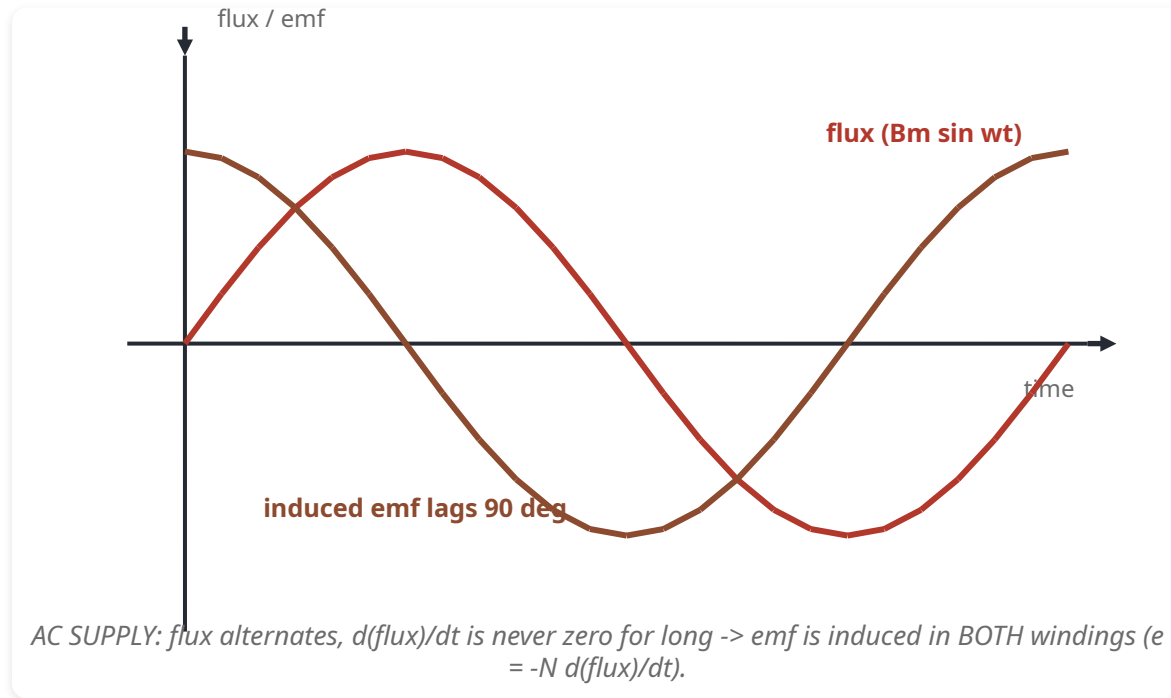
1. AC supply to primary N1 -> alternating current i_1 .
2. Alternating MMF = $N1 \times i_1$ in the core.
3. Alternating mutual flux B_m set up, confined to the low-reluctance iron path.
4. The SAME flux links the secondary N2 (mutual induction).
5. Faraday: emf is induced in N2 because $d(\text{flux})/dt$ is non-zero: $E2 = 4.44 f N2 B_m A$.
6. Close the secondary on a load -> current flows -> power delivered at transformed voltage.

STEP-UP / STEP-DOWN

$N2 > N1 \rightarrow V2 > V1$ (step-up). $N2 < N1 \rightarrow V2 < V1$ (step-down). The transformation ratio $k = N2/N1 = E2/E1 = I1/I2$.

Why a transformer needs AC (A1-16 answer)

ROTE 2 | THE WHY



A1-16 MODEL ANSWER A transformer works on mutual induction, which demands a changing flux. With DC, flux is constant, so the induced emf $e = -N d(\text{flux})/dt$ is zero, there is no back-emf to oppose the supply, and the primary winding draws a current limited only by its own resistance (V/R). That current is very large and overheats/destroys the winding. Also the core saturates. Conclusion: transformers are rated and used with AC only.

Losses in a transformer

ROTE 3 | LOSSES

TOTAL LOSS = copper + iron (+ stray)

COPPER LOSS (winding, load-dependent)

$P_{cu} = I_1^2 R_1 + I_2^2 R_2 = I^2 R$ (varies with the square of load current)

IRON / CORE LOSS (constant at fixed V and f)

(a) Hysteresis loss: energy spent re-orienting domains every cycle.

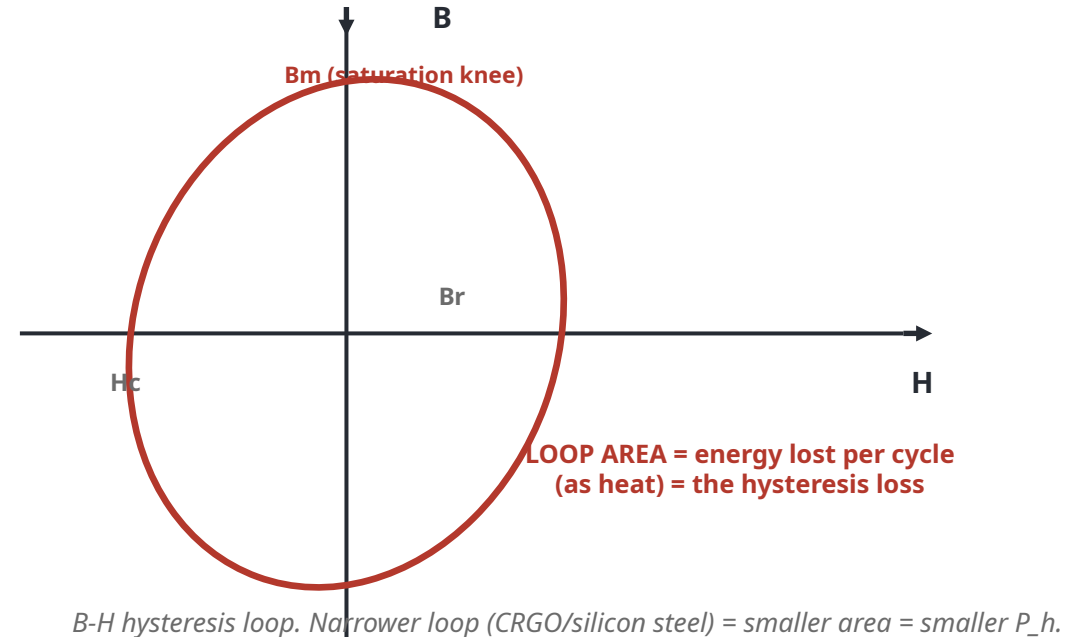
$$P_h = k_h \times f \times B_m^{1.6}$$

(b) Eddy-current loss: circulating currents induced IN the core itself.

$$P_e = k_e \times f^2 \times B_m^2 \times t^2$$

STRAY LOSSES: small leakage-flux losses in tank/frame; conventionally a small % of output.

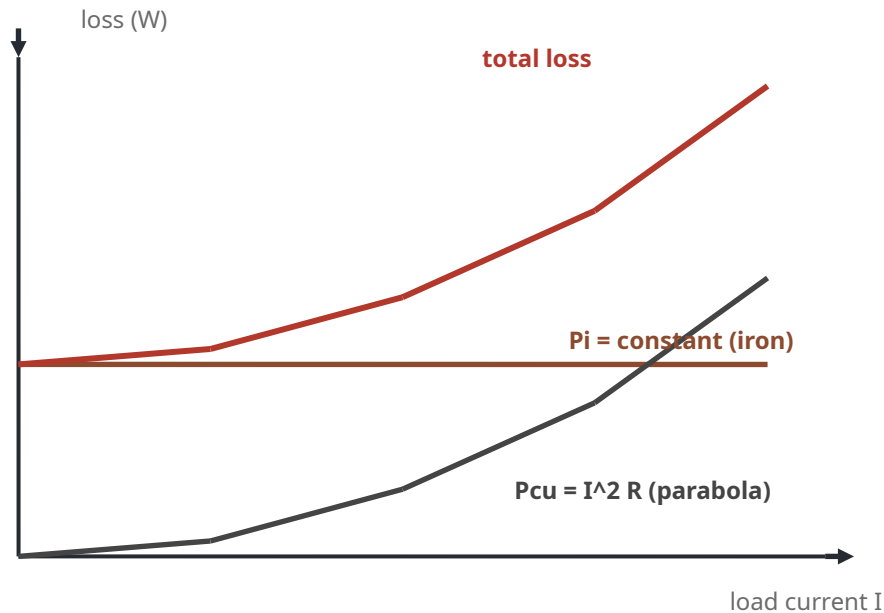
NO mechanical losses: the transformer is a STATIC machine (no rotor, no bearings).



One-line conclusion: copper follows the load current squared; iron follows voltage and frequency and stays flat. Static machine: no mechanical loss.

Reducing the losses, and why transformers are rated in kVA

ROTE 3 | REDUCE + RATING



Losses vs load: iron flat, copper quadratic. Efficiency peaks where $P_{cu} = P_i$ (that is N07).

HOW TO REDUCE (A1-20 marks live here)

- Hysteresis loss: use CRGO / silicon steel (narrow B-H loop, less energy per cycle).
- Eddy-current loss: laminated core (insulated sheets 0.27-0.35 mm). Loss falls with t^2 .
- Copper loss: use thicker conductors (lower R); keep windings short (lower mean turn length).
- Stray: careful winding/flux-path design + magnetic shielding.

WHY kVA AND NOT kW (A1-19)

Core loss depends mainly on VOLTAGE. Copper loss depends mainly on CURRENT. Neither loss knows the load power factor. But $kW = kVA \times pf$, so if the manufacturer rated in kW, the safe kVA would depend on the customer's pf. Rating in kVA keeps the temperature rise safe for ANY pf the user connects.

One-line conclusion: iron fixed by V and f, copper by I squared, so the honest rating is kVA.

A1-01: the model answer you reproduce

WORKED | FULL ANSWER

Q: Explain the construction and working of a single phase transformer.

Construction. A single phase transformer is a static (non-rotating) electrical machine that transfers AC power from one circuit to another at the same frequency but a different voltage, by mutual induction. Its magnetic circuit is a core built of laminated CRGO (cold-rolled grain-oriented) steel sheets, typically 0.27-0.35 mm thick and insulated from each other; the laminations suppress eddy currents and the grain-oriented steel reduces hysteresis loss. The core has two limbs (core type: winding surrounds the core) or three limbs with the winding on the centre limb (shell type: core surrounds the winding). A yoke joins the limbs so the flux has a closed, low-reluctance path and the assembly is mechanically rigid. Two windings sit concentrically on the limbs: the primary and secondary. The LV winding (few turns of thick wire) is placed next to the core and the HV winding (many turns of thin wire) over it, so the insulation volume is minimum. Windings, core and leads live in a sheet-steel tank filled with transformer oil (insulation + cooling), with a conservator for oil expansion, bushings to bring the terminals out, and a silica-gel breather to keep the air dry.

Working. When an AC supply is given to the primary (N_1 turns), an alternating current flows and produces an alternating magnetomotive force $N_1 \times i_1$. This sets up an alternating mutual flux in the core, which is confined to the low-reluctance iron path and therefore links the secondary (N_2 turns) also. Because the flux is alternating, by Faraday's law an emf is induced in each winding, $e = -N \frac{d(\text{flux})}{dt}$; in rms terms $E_1 = 4.44 f N_1 B_m A$ and $E_2 = 4.44 f N_2 B_m A$ (the $4.44 f N B_m A$ law is derived in N02). With the secondary circuit closed on a load, the induced E_2 drives a current and power is delivered to the load. The voltage is transformed in the ratio $E_1/E_2 = N_1/N_2 = 1/k$, while the frequency is unchanged.

Conclusion. One alternating flux, two electrically separate circuits: voltage and current transform by the turns ratio, power (minus losses) crosses unchanged, frequency stays the same.

Label checklist if you redraw the figure: yoke, limbs, LV winding, HV winding, insulation, tank + oil, conservator, bushings, flux path arrows.

A1-20 model answer + closed-book check

A1-20: List the losses and how to reduce them.

A transformer has two main losses. (1) Copper loss, the I^2R heating of both windings, which varies with the square of load current. (2) Iron (core) loss, constant at fixed voltage and frequency, in two parts: hysteresis loss (energy spent reversing domains every cycle, $P_h = k_h f B_m^{1.6}$) and eddy-current loss (circulating currents induced in the core metal, $P_e = k_e f^2 B_m^2 t^2$). Small stray losses also appear in tank/frame from leakage flux. Reduction: hysteresis by CRGO/silicon steel (narrow B-H loop), eddy by laminating the core (loss falls with t^2), copper by thicker conductors and shorter mean turn, stray by careful design and shielding. Because core loss depends on voltage and copper loss on current, neither follows the power factor, so the transformer is rated in kVA, not kW.

ONE-PAGE MEMORY DUMP (write cold)

core: laminated CRGO, gives the flux path

yoke: closes flux path + supports

LV: thick, few turns, next to limb

HV: thin, many turns, outermost

core type: winding around core (HV)

shell type: core around winding (LV)

working: AC $\rightarrow$ MMF $\rightarrow$ flux $\rightarrow$ Faraday $\rightarrow$ E2

$E = 4.44 f N B_m A$; $k = N_2/N_1 = E_2/E_1 = I_1/I_2$

losses: $P_{cu} = I^2R$; $P_h = k_h f B_m^{1.6}$; $P_e = k_e f^2 B_m^2 t^2$

static machine $\rightarrow$ NO mechanical loss

rating: kVA (losses track V and I, not pf)

CLOSED-BOOK CHECK (cover this slide, answer out loud)

1. Name every part of a single phase transformer and its function in one line each.
2. Draw and label core type and shell type, and say which suits HV and why.
3. Give the 6-step working sequence ending at the load current.
4. Why does a transformer fail on DC? Give the full argument.
5. List all losses, two reduction methods per iron loss, and one sentence on why the rating is kVA.

ANSWER STRUCTURE WHEN PANIC HITS define $\rightarrow$ draw and label the diagram $\rightarrow$ state the governing formula or sequence $\rightarrow$ substitute/speak through the steps $\rightarrow$ finish with a one-line conclusion. Never leave it blank.